

Affective visualization graphs

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Table of contents

Setting up	1
Framing my question	2
Cleaning and wrangling data	2
Fitting the model	2
Checking diagnostics	3
Looking at the model coefficients	4
Looking at the model predictions	5
Visualizing model predictions	6
Describing the model	7

Setting up

```
# read in packages
library(tidyverse) # general use
library(here) # file organization -- dont need unless sort into folders
library(janitor) # cleaning data frames
library(DHARMA) # for diagnostics
library(ggeffects) # for ggpredict
library(flextable) # for flextable
```

```
# read in personal data to object from csv
my_data <- read_csv("envs193ds-personal-data.csv")
```

Framing my question

My central question is: How does sleep duration the night prior influence the probability of me going on a run the next day?

Variables

- Predictor: sleep duration the previous night (hours)
- Response: probability of going on a run (1/0) (1 = yes; 0 = no run)

Cleaning and wrangling data

```
# cleaned column names
my_data <- my_data |>
  clean_names()

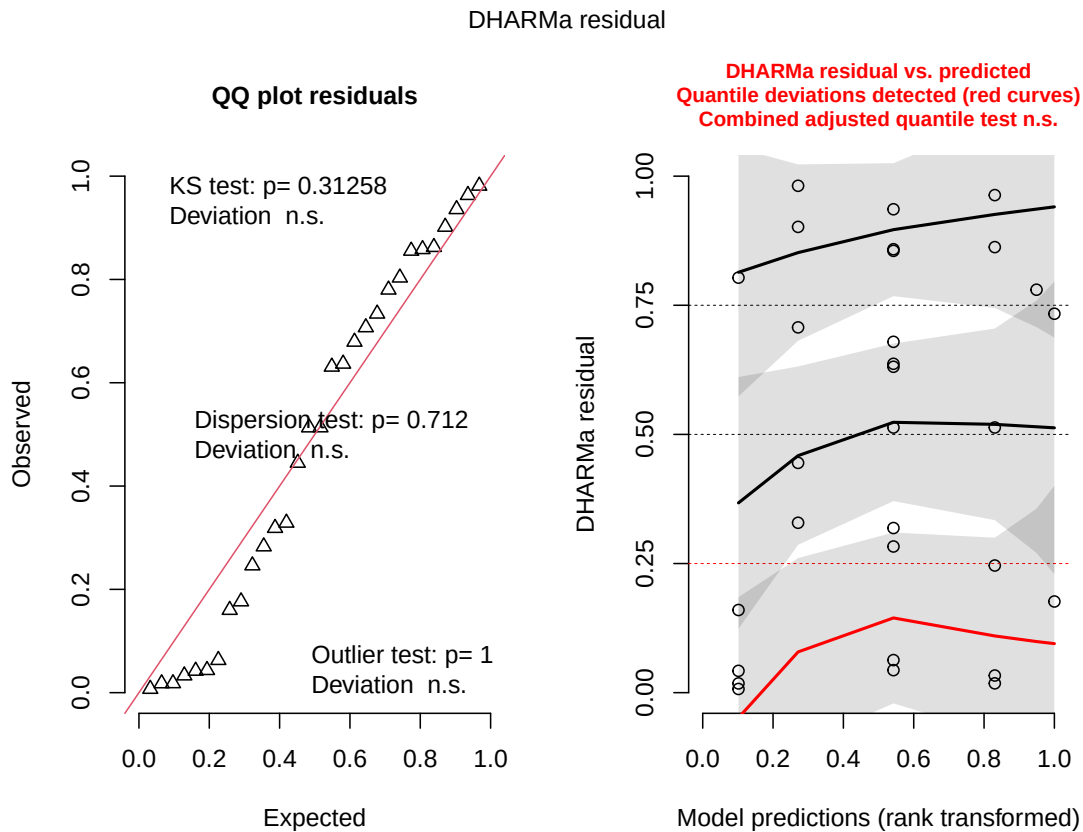
# created new column: 1 if workout was a run, 0 if not
my_data <- my_data |>
  mutate(is_run = case_when(
    workout_type == "Run" ~ 1, # run = 1
    .default = 0))             # all other workout types = 0
```

Fitting the model

```
# fitted generalized linear model for running
# response: whether or not workout was a run (binary: 1/0)
# predictor: sleep the previous night (hours)
run_mod <- glm(
  # formula: binary run outcome ~ sleep hours
  is_run ~ sleep_hours,
  # data: my_data data frame
  data = my_data,
  # binomial error distribution for binary response
  family = binomial)
```

Checking diagnostics

```
# simulated residuals from GLM for diagnostic plots  
plot(simulateResiduals(run_mod))
```



Info from left plot (QQ plot of residuals):

- KS test: do my residuals follow a uniform distribution? → generally yes (CHECK)
- Dispersion test: are your residuals more dispersed than expected from a uniform distribution? → no
- Outlier test: are there outliers in your data set that could be influencing model coefficients or predictions? → no

Info from right plot (residuals vs predicted values):

- By default, R generates the reference lines for quartiles: 25, 50, 75

- Not that informative but look at patterns within residuals across the range → DHARMA diagnostics flag issues in 25th quartile due to small sample size, not because the model is wrong.

Looking at the model coefficients

```
# displayed model summary
summary(run_mod)
```

Call:

```
glm(formula = is_run ~ sleep_hours, family = binomial, data = my_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-8.7011	3.4964	-2.489	0.0128 *
sleep_hours	1.0873	0.4712	2.307	0.0210 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36.652 on 29 degrees of freedom
 Residual deviance: 28.046 on 28 degrees of freedom
 AIC: 32.046

Number of Fisher Scoring iterations: 5

```
# exponentiated coefficients to get odds ratios
# e.g. exp(coef) = odds ratio for 1 hr increase in sleep
exp(coef(run_mod))
```

```
(Intercept)  sleep_hours
0.0001663954 2.9663093920
```

```
# exponentiated confidence intervals
exp(confint(run_mod))
```

	2.5 %	97.5 %
(Intercept)	3.108829e-08	0.04889274
sleep_hours	1.372042e+00	9.39665076

Note: CHECK. for every 1 hour increase in sleep, the the odds of going on a run increase by a factor of 2.97 (95% CI: [X, X], $p = X$, $= 0.05$).

Looking at the model predictions

- using probabilities to determine chance of getting certain outcome
- line represents the probability of 1 (going on a run) actually occurring

```
# what is the probilty of going on a run after sleeping for 10 hours?
ggpredict(run_mod,
  terms = "sleep_hours [10]")
```

Predicted probabilities of is_run

sleep_hours	Predicted	95% CI
10	0.90	0.39, 0.99

```
# what is the probilty of going on a run after sleeping for 4 hours?
ggpredict(run_mod,
  terms = "sleep_hours [4]")
```

Predicted probabilities of is_run

sleep_hours	Predicted	95% CI
4	0.01	0.00, 0.25

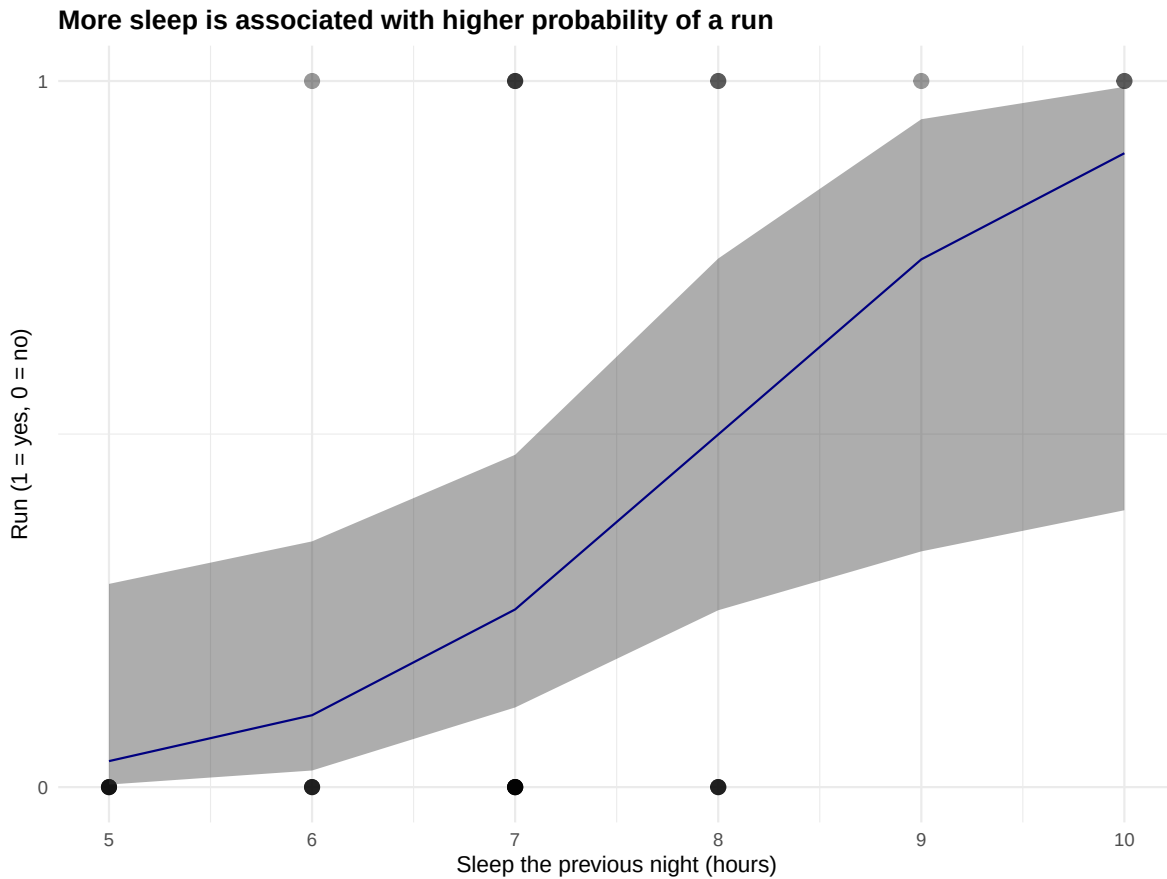
Note:

- If I slept for 10 hours, the probability of going on a run the next day is 0.90 (95% CI [0.39, 0.99]).
- If I slept for 4 hours, the probability of going on a run the next day is 0.01 (95% CI [0.00, 0.25]).

Visualizing model predictions

```
# generated model predictions across range of sleep values
mod_preds <- ggpredict(run_mod,
                      terms = "sleep_hours [5:10 by = 1]")

# base layer: ggplot with raw data
# data frame: my_data
ggplot(my_data,
       aes(x = sleep_hours,
           y = is_run)) +
  # first layer: raw observations
  geom_point(size = 3,
            alpha = 0.4) +
  # second layer: confidence interval ribbon from model predictions
  geom_ribbon(data = mod_preds,
            aes(x = x,
                y = predicted,
                ymin = conf.low,
                ymax = conf.high),
            alpha = 0.4) +
  # third layer: logistic curve from model predictions
  geom_line(data = mod_preds,
           aes(x = x,
               y = predicted),
           # custom color
           color = "navyblue") +
  # constrained y-axis to 0 and 1
  scale_y_continuous(limits = c(0, 1),
                    breaks = c(0, 1)) +
  # relabeled axes and added title
  labs(title = "More sleep is associated with higher probability of a run",
       x = "Sleep the previous night (hours)",
       y = "Run (1 = yes, 0 = no)") +
  # changed theme from default
  theme_minimal() +
  # bolded title
  theme(plot.title = element_text(face = "bold"))
```



Describing the model

```
# displaying model in table
as_flextable(run_mod)
```

	Estimate	Standard Error	z value	Pr(> z)
(Intercept)	-8.701	3.496	-2.489	0.0128 *
sleep_hours	1.087	0.471	2.307	0.0210 *

Estimate	Standard Error	z value	Pr(> z)
<i>Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '.' < 0.05</i>			

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36.65 on 29 degrees of freedom

Residual deviance: 28.05 on 28 degrees of freedom